

Week 3 Progress Report

Intern Name: Khadija Ayub

Project: Employee Management System Dashboard

Week: Week 3

Objective

Enhance the Employee Management Dashboard by integrating employee data from a mock REST API and improving user experience through search, sorting, pagination, loading skeletons, and robust state handling.

Tasks Completed

- Integrated the **DummyJSON Users API** using the Fetch API to retrieve employee data.
- Transformed the API response into the application's employee model before rendering.
- Implemented **loading skeletons** to improve the user experience while data is being fetched.
- Added **error handling** to display meaningful messages when API requests fail.
- Implemented an **empty state** message when no employee records match the search criteria.
- Developed **search** functionality to filter employees by their full name in real time.
- Implemented client-side **sorting** for Full Name, Department, and Salary with ascending and descending order.
- Implemented client-side **pagination** with Previous, Next, and page number navigation.
- Reset pagination automatically when the search query changes for improved usability.
- Updated the employee form to correctly display API joining dates using the HTML date input format.
- Improved the responsiveness and usability of the employee table.

React Concepts Practiced

useState, useEffect, Conditional Rendering, Component Communication, Props, Fetch API Integration, State Management, Client-side Searching, Sorting, Pagination.

JavaScript Concepts Practiced

Fetch API, Async/Await, Try-Catch-Finally, map(), filter(), sort(), slice(), Array.from(), Spread Operator, Arrow Functions, Ternary Operator..

Challenges Faced

Transforming API data into the required employee model, handling asynchronous data fetching, implementing efficient client-side sorting and pagination together, formatting dates correctly for HTML date inputs, and managing multiple UI states such as loading, empty, and error conditions.

Outcome

Successfully integrated a mock REST API into the Employee Management Dashboard and enhanced the application with loading skeletons, client-side search, sorting, pagination, empty state handling, and error handling, resulting in a more interactive and user-friendly experience.

Next Week Goals

Extract reusable UI components such as buttons, inputs, cards, badges, and table controls, make them configurable through props, and improve component reusability and documentation across the project.